A SQL PORTFOLIO PROJECT

Mortgage lending in New York

Every mortgage application filed in New York State over four years, as the lenders reported it. Modelled in PostgreSQL, analysed in forty-two queries, and runnable in your browser without installing anything.

1,596,676
applications decided

951,371
loans originated

$371.4bn
disbursed

Samuel Babajide · HMDA, New York State, 2022 to 2025

swipe →

THE MARKET

Four years, year on year

YEAR	APPLICATIONS	ORIGINATED	DISBURSED	APPROVAL	DENIAL
2022	500,553	307,252	$128.5bn	64.39%	18.69%
2023	346,943	203,072	$72.7bn	61.72%	21.68%
2024	355,249	208,814	$76.9bn	61.94%	21.02%
2025	393,931	232,233	$93.3bn	62.71%	19.63%
All four	**1,596,676**	**951,371**	**$371.4bn**	**62.85%**	**20.09%**

Applications fell 30.7 percent from 2022 to 2023 and have recovered 13.5 percent since. Approval bottomed in 2023 at 61.72 percent and denial peaked the same year at 21.68. 2022 is a three-year file and 2025 is provisional, so the four vintages are not equally complete.

THE TYPICAL LOAN

What a New York mortgage actually looks like

$127,000	$475,000	$245,000
MEDIAN APPLICANT INCOME	MEDIAN PROPERTY VALUE	MEDIAN LOAN AMOUNT

75.00%	1.96×	6.281%
MEDIAN LOAN TO VALUE	MEDIAN LOAN TO INCOME	MEDIAN INTEREST RATE

Across the 951,371 loans that were actually made. Every median uses **percentile_disc**, which returns a value some borrower really had, rather than interpolating between the two middle rows and reporting a figure nobody was charged. That choice is also what makes these numbers identical in PostgreSQL and in the browser.

COVERAGE

What the register covers

4	1,168	62	5,281
YEARS	LENDERS	OF 62 COUNTIES	CENSUS TRACTS

4	6	4	9
LOAN TYPES	LOAN PURPOSES	SEX VALUES	RACE VALUES

MOST EXPENSIVE COUNTY	MEDIAN VALUE	LOANS
New York County	$1,355,000	35,463
Kings County	$1,165,000	53,975
Queens County	$805,000	57,169

All 62 New York counties appear. The raw column returns 66, because it also carries an **UNKNOWN** value and three out-of-state codes filed against New York property. Counting those would have claimed four counties that do not exist.

APPLICANT DEMOGRAPHICS

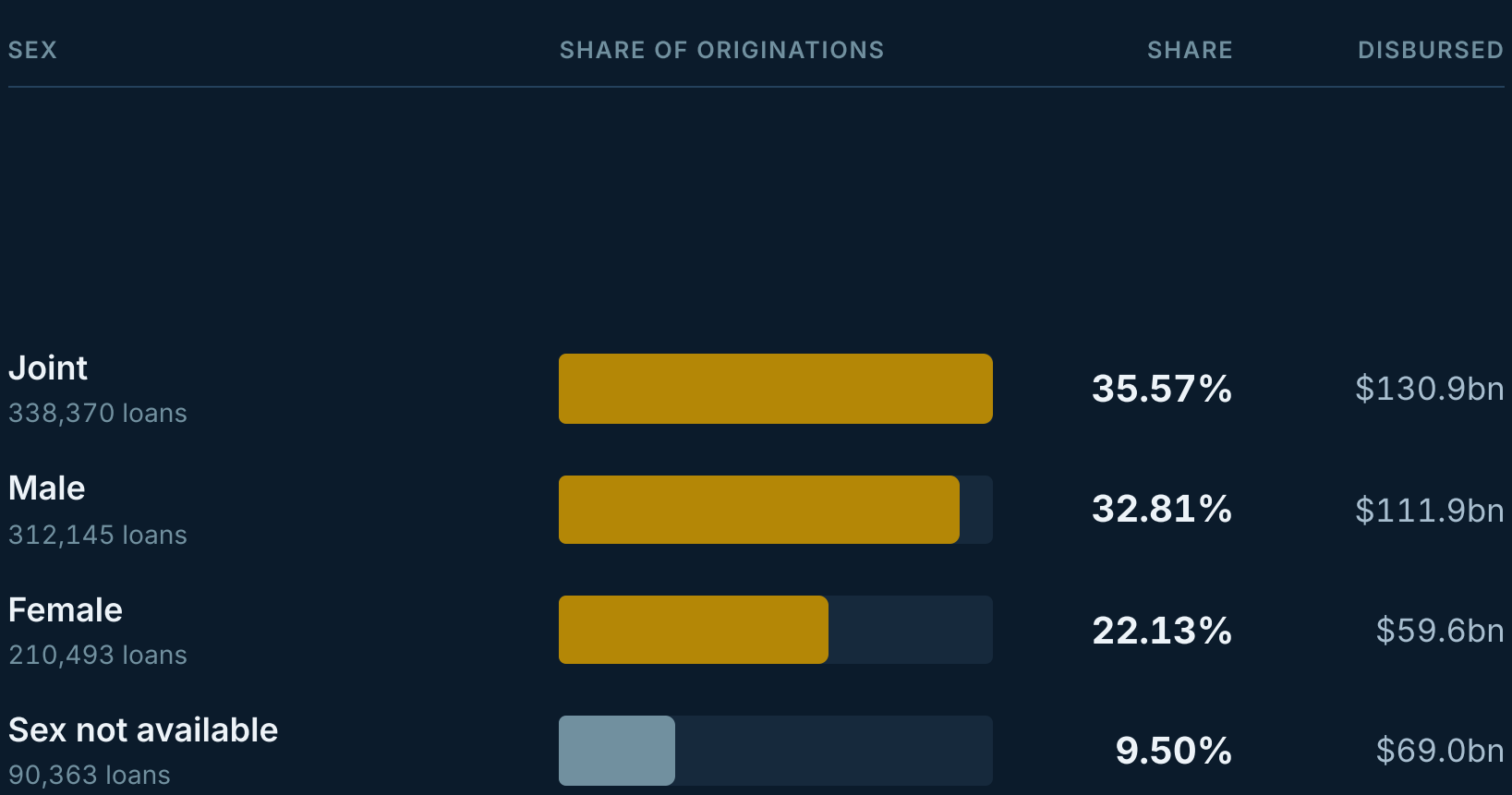
Originated loans by applicant race

RACE	SHARE OF ORIGINATIONS	SHARE	DISBURSED
White 621,461 loans		65.32%	$198.2bn
Race not available 161,122 loans		16.94%	$98.9bn
Asian 87,024 loans		9.15%	$44.1bn
Black or African American 58,130 loans		6.11%	$19.6bn
Joint 17,886 loans		1.88%	$8.9bn
Four smaller categories 5,748 loans		0.60%	$1.7bn

The regulator's categories are never dropped here, including "not available". It is 16.94 percent of originations and 26.6 percent of the dollars, so removing it would move every other share on this slide without saying so. The six rows sum to 951,371 loans exactly.

APPLICANT DEMOGRAPHICS

Originated loans by applicant sex



Look at the last row twice. "Sex not available" is 9.50 percent of the loans and **18.57 percent of the dollars**, the only row where the two diverge sharply, on a median loan of **$335,000** against $265,000 for joint applicants. These are largely applications filed by a company rather than a household, which is why the field is blank, and it is the clearest reason not to delete a "missing" category.

DENIAL ANALYSIS

The five reasons lenders write down most

REASON CITED	SHARE OF ALL CITATIONS	SHARE	CITATIONS
Debt-to-income ratio		**32.87%**	135,207
Credit history		**21.71%**	89,302
Collateral		**17.45%**	71,778
Other		**11.95%**	49,153
Credit application incomplete		**6.00%**	24,663

The unit is a citation, not an application. HMDA lets a lender give up to four reasons, so 320,762 denied applications carry **411,362 citations**. Five more reasons share the remaining 10.02 percent. Debt-to-income is cited more often than collateral and "other" combined.

WHAT THE DATA RECOMMENDS

Three findings a borrower can act on

1 Lower is not better, and the reason matters

Under 20 percent debt-to-income denies at **24.49 percent**, worse than every band from 20 to 49, where the rate is flat near **13**. That bottom is not cautious borrowers: it is small second-lien business. The threshold is a line, not a direction. Get under 50. Below 20 buys nothing.

2 Which lender you ask is worth 15 times

Same product in every row, conventional first-lien home purchase, 72 lenders with 2,000 or more such applications. Denial runs **1.14 to 31.57 percent**. SEFCU denies **1.14** on **more leveraged, lower-income** applicants than Bank of America, which denies **17.76**. Most borrowers approach one lender.

3 Borrowing against a home you own is a different market

Home improvement denies at **35.13 percent** and other purpose at **35.84**, against **10.78** for a purchase. With **493 and 479** lenders active rather than 934. Three times as hard, with half the choice.

WHAT THIS CANNOT SAY

The limits, stated rather than buried

HMDA records what lenders decided and the reason they wrote down. It holds no credit score, no assets and no employment record. So every figure in this carousel is a difference in **outcomes**, and nothing here isolates the decision that produced it. Three conventions decide most of the numbers.

Purchased loans are excluded everywhere

Action code 6 records a loan a bank bought after another institution made it. The buyer made no decision, so the row belongs in neither the numerator nor the denominator of a rate. 158,743 rows excluded, leaving 1,596,676.

An application ends in one of four ways, and the four sum to 100

Approval, denial, file closed for incompleteness, withdrawal. Reporting all four is what makes several findings visible at all: a lender can reach a low denial rate by deciding well or by losing applications before a decision.

Categories the regulator defines are never dropped

Including the codes for "not available" and "not provided". They are not groups of people, and removing them would change every share in every table without saying so.

DON'T TAKE MY WORD FOR IT

Run the queries yourself

42
queries, each one printed beside its result

2
engines, agreeing cell by cell

0
installs, accounts or servers

1.75 million rows of Parquet, queried in your own browser tab by DuckDB compiled to WebAssembly. Nothing is installed, no account exists and no data leaves your machine. Every figure in this carousel is one click from the query that produced it, and an automated parity test runs all forty-two against PostgreSQL 18 and DuckDB and fails on any difference. **Forty-two of forty-two agree.**

→ The playground | → The data dictionary | → The ERD | → The repository

samuelbabajide.github.io/us-home-mortgage-disclosure-act-new-york

Built with PostgreSQL 18 · dimensional modelling · data quality assertions · query optimisation · Python · DuckDB WASM · Parquet · GitHub Pages

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